浙江大學

Lab Practice Report for Undergraduate

Name:

College: ZheJiang University

Department: Science

Major: Marine Science

Student NO.:

Instructor: Yanan Di

Date: 2024.10.10

Lab Practice Report of Zhejiang University

（此页可在 http://bksy.zju.edu.cn/office/下载）

Course Name： Marine biology Type of Practice： dissection

Name of Practice： Quantitative confirmation of microalgae

Student Name： Major： Marine biology

Student Number：

Team Memeber： Zhiang Liu

Instructor： Yanan Di

Practice Location： 241 Practice Date: 2024.11.7

装订线

1. Aims and Requirement of the Practice (Compulsory)

(1) learn about mollusca biology and complete a live or a virtual dissection of mussels and snail

(2) Identify the external anatomy of typical mollusca;describe the function of important external features.

(3) Identify the major internal organs of mussels;describe their biological functions

(4) Demonstrate dissection skills

2. Equipment and materials

1. organisms:mussels and snail
2. Dissection set
3. Ice box and debris ice
4. 1.5ml centrifuge tubes and racks
5. Liquid nitrogen and container
6. Marker pens
7. Gloves,plastic bags paper towels.

3. Procedures

1. Select one organism and put it on ice.Observe it to identify its gender and basic body structure.Measure the size and make record.

2. Dissection of mussels

3. Homogenziation and Preservation of tissue samples

(1)Cut muscle sample and digest sample(no bigger than 0.27 cm^3,3 replicates)

(2)Put 1.5ml centrifuge tubes,add sample into it.Add 500µl PBS into each tube.Homogenize it 1min with homogenizer 1min.Then add 500µl PBS again and homogenization 10s.

(3)Put tubs into centrifuge,spin at 4℃,12000g for 5min.When it finish,collected supernatant and transport it to another clear 1.5ml tube.Mark it and sent it to -20℃

Refrigerator

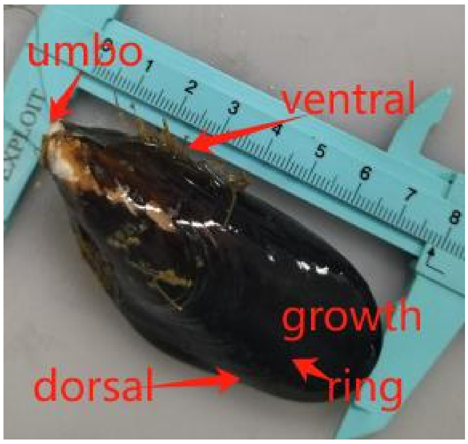
(4)Clean up everything you used.Fold all materials into paper towels and put into plastic bags.Rinse and sanitize dissection equipment.

4. Questions

(1) take picture of your samples;identify the external structures and functions

Mussels: Size（long x wide x high）=7.81 cm x 3.82cm x 2.53cm

Have 42 growth rings total



Snails: Size（long x wide x high）=2.32 cm x 1.81cm x 1.93cm

Define;Lymnaeidae(Uncertained)

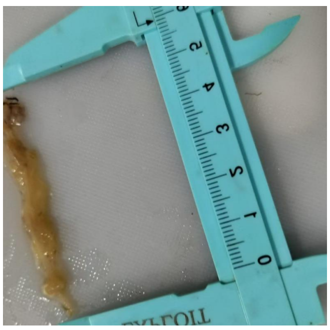
(2) Sample quality table

Sample number	tube quality（g）	Sample and tube quality（g）	Sample quality（g）
M1	0.8378	0.9829	0.1451
M2	0.8398	0.9940	0.1542
M3	0.8263	0.9797	0.1534
D1	0.8481	0.9916	0.1435
D2	0.8447	0.9885	0.1438
D3	0.8156	0.9841	0.1685

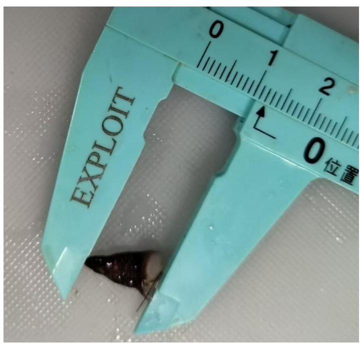
(3) Dissect the organisms and make forms to show different internal structures of mussel and snail

Mussel:

Gills:



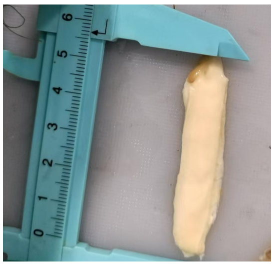
foots:



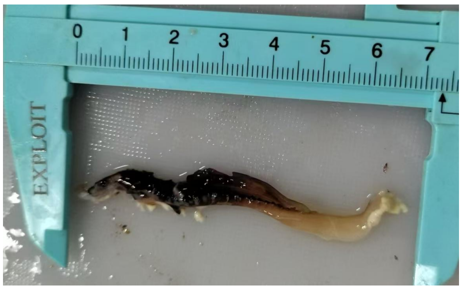
Adductor Muscle:

digestive gland:

Mantle（male）:



siphonal edge:



Snail:

Heart:

operculum:

Stomach:

intestine

Foot:

radula and nose:

(4) Discuss the internal structures and their functions

structures	functions
foot	Helps with movement
Stomach	Digestive sites
Heart	Blood supply
intestine	Absorb nutrients from food
Gills	Help breathe
digestive gland	Secrete digestive enzymes
Adductor Muscle	control the opening and shutting of shells
mantle	Secretes shells and protects internal organs
siphonal edge	Breathe and filter food
nose	Exposure to odor stimuli
radula	Physically cut food
operculum	Protects internal organs

装订线